

Homework 1

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Exercise 1

Let S be a finite set of positive integers. In the S -nim game, we have a pile of n chips, and two players take turns removing chips from the pile. The number of chips that a player removes in a turn must be in the set S . The player who cannot move loses. Determine the pile sizes for which the starting player has a winning strategy in $\{2, 4, 7\}$ -nim.

Solution: The positions starting from 0 have the following N and P assignments:

$P, P, N, N, N, N, P, N, N, P, P, N, N, P, P, N, N, P, \dots$

We can see a pattern here:

$P, P, N, N, N, N, | (P, N, N), (P, N, N), (P, N, N), (P, N, N), \dots$

From this we can determine the rule for which positions are N or P :

$$x \in P \iff \begin{cases} x < 6 \text{ and } x \in \{0, 1\} \\ x \geq 6 \text{ and } (x - 6) \equiv 0 \pmod{3} \end{cases}$$

For cases $x < 6$ we can check by hand, for cases $x \geq 6$ we can see that if we are in a position $(x - 6) \not\equiv 0 \pmod{3}$ then we can always jump to a position that is in P .

Likewise if we are in a position which is $(x - 6) \equiv 0 \pmod{3}$ then there is not jump we can make that will take us back to a position in P . □

Exercise 2

Consider the following two-player game. There are two piles of chips, of size n and m respectively. The two players take turns. In each turn, the player discards one of the piles, and divides the remaining pile arbitrarily into two non-empty piles. The player who cannot move loses (this happens when both piles contain a single chip). Determine the (n, m) values for which the first player has a winning strategy.

Solution: For figuring out the solution it is helpful to write out the table for the first 7×7 positions:

$$\begin{pmatrix} & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \end{matrix} & \begin{matrix} P \\ N \\ P \\ N \\ P \\ N \\ P \end{matrix} & \begin{matrix} N \\ N \\ N \\ N \\ N \\ N \\ N \end{matrix} & \begin{matrix} P \\ N \\ P \\ N \\ P \\ N \\ P \end{matrix} & \begin{matrix} N \\ N \\ N \\ N \\ N \\ N \\ N \end{matrix} & \begin{matrix} P \\ N \\ P \\ N \\ P \\ N \\ P \end{matrix} & \begin{matrix} N \\ N \\ N \\ N \\ N \\ N \\ N \end{matrix} & \begin{matrix} P \\ N \\ P \\ N \\ P \\ N \\ P \end{matrix} \end{pmatrix}$$

Clearly the matrix is symmetric because position (n, m) is equivalent to the position (m, n) , thus we do not write out the full matrix, only the upper right half.

From the position matrix we make the following conjecture:

$$(n, m) \in P \iff n \equiv 1 \pmod{2} \quad \text{and} \quad m \equiv 1 \pmod{2}$$

This conjecture can be proven by induction.

Suppose the conjecture is correct up until some $n_0, m_0 \in \mathbb{N}$.

Given any other position (n, m) if $n \equiv 0 \pmod{2}$ then we can split n into $(1, n - 1)$ where this position will be in P both 1 and $n - 1$ are odd, concluding $(n, m) \in N$.

If both n and m are odd, then no matter which we choose, we cannot split an odd number as a sum of two odd numbers. □